

ICAI AICA LEVEL-2 · INDIVIDUAL CAPSTONE PROJECT · BATCH 88

GST Blocked Credit Checker

A rule-based decision-support tool for Input Tax Credit eligibility under Section 17(5) of the CGST Act, 2017 — with an independent Reverse Charge Mechanism alert.

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01 • INTRODUCTION

"Can I claim ITC on this?" — a question every CA answers, every day

Why it matters

Section 17(5) of the CGST Act blocks Input Tax Credit on roughly sixteen categories of expense — each with its own exceptions that are easy to misremember under time pressure.

What was built

An AICA Level-2 capstone project: a Streamlit web app and command-line tool that gives an instant, clause-referenced verdict — built on one shared rules engine in Python.

02 • PROBLEM STATEMENT

Sixteen blocked-credit categories — each with its own exceptions

17(5)(a) • Motor Vehicles

Blocked in general — but a cab or driving-school operator can still claim it.

17(5)(b)(i) • Food & Catering

Blocked in general — but eligible when obligatory under any law for the employer.

17(5)(b)(iii) • Club Membership

Squarely blocked, with no employee-benefit carve-out under this clause.

17(5)(c)/(d) • Construction

Amended by the Finance Act, 2025 — "plant or machinery" now reads "plant and machinery", retrospectively.

Getting the exception wrong means an avoidable disallowance — or legitimate credit left unclaimed.

03 • THE SOLUTION

One rules engine, three ways to reach it



Quick Check

Type one expense description. If the clause needs a fact — seating capacity, statutory obligation — the tool asks a follow-up before giving the verdict.



Bulk Check

Upload a whole expense ledger in Excel. Get back a colour-coded, clause-referenced verdict for every row, with a Needs Review flag where a fact is missing.



Invoice Check

Upload a photo or PDF invoice. The tool reads it — via AI, PDF text, or offline OCR — then runs the extracted text through the same engine.

All three tabs share exactly one rules engine — the legal logic is defined once, so it can't drift.

03 • THE SOLUTION (CONTD.)

Two independent questions, one description

SECTION 17(5)

Can I claim ITC on it?

The primary verdict — eligible, blocked, or needs review — with the exact clause and reasoning.

SECTION 9(3)/9(4) — RCM

Who is liable to pay the GST?

A separate, independent alert — shown alongside the verdict, never instead of it — for supplies commonly taxed under reverse charge.

Example: legal fees from an advocate are RCM-liable (you self-invoice and pay the GST) **and** fully ITC-eligible at the same time — both answers matter.

04 · TOOLS & TECHNOLOGY

Built with Python, Streamlit, and Claude

Layer	Technology	Role in the tool
Core logic	Python	Rules engine, RCM list, keyword matching
Web app	Streamlit	Quick Check, Bulk Check, Invoice Check tabs
Bulk reports	pandas, openpyxl	Excel ledger read-in, colour-coded report output
Invoice reading	PyMuPDF, Tesseract OCR	PDF text layer, offline OCR fallback
AI / LLM	Anthropic Claude (claude-sonnet-4-5)	Reads invoice photos/PDFs; plain-language explanation of a verdict

Claude is optional throughout — with no API key, Invoice Check still works fully offline, and no verdict ever depends on the AI call succeeding.

05 • KEY TAKEAWAYS

- Rule-based and explainable — every verdict cites the exact clause, never a black box.
- One engine, zero drift — Quick Check, Bulk Check, and Invoice Check all read from the same Section 17(5) rules.
- RCM Alert flags reverse-charge liability independently, alongside the ITC verdict.
- A decision-support aid, not a substitute for professional judgement — every result should be verified against the Act.

Thank you

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